PrepAIred: A Guardrailed Technical-Interview Practice System Implementation Audit and Scoped Evaluation

Anonymous authors
Affiliation withheld

Abstract—Technical-interview practice systems face a tension between static question banks, which cannot adapt to a candidate's performance, and language-model-driven interviewers, whose grading and adaptation logic risks being entangled with the same model that generates conversational text. We report a full implementation audit and evidence-scoped evaluation of PrepAIred, a running system that separates these concerns: a deterministic evaluator combines sentence-embedding similarity, retrieval-grounded concept coverage, and a partially fine-tuned CrossEncoder reasoning score into a verbal-answer grade; an independent acoustic-and-linguistic pipeline estimates communication confidence; a PPO policy proposes difficulty changes subject to a deterministic guardrail layer and a heuristic fallback; and a local, quantized language model generates follow-up questions and feedback text only after a separate, evidence-gated policy has decided whether and what to ask, never the score or difficulty. Auditing the implementation against its own design documentation surfaced five previously undocumented discrepancies, including a mistake-penalty applied twice to the same score and a deployed reinforcement-learning checkpoint distinct from the one any simulation result describes. On a 64-answer exploratory human benchmark, the evaluator's full composite reaches Spearman $\rho = 0.38$, below two simpler configurations on the same data; a confirmatory re-run shows the fine-tuned CrossEncoder is not established to outperform a trivial answer-length baseline. Across two non-overlapping simulated persona populations under a shared guardrail layer, the adaptive controller's benefit over simpler policies is population-dependent and unevaluated for the checkpoint actually deployed. No real-user effectiveness, learning-gain, or fairness claim is made.

Index Terms—technical interview practice, automated answer evaluation, CrossEncoder re-ranking, adaptive difficulty, reinforcement-learning guardrails, speech confidence estimation, LLM authority boundaries

I. Introduction

Practicing for a technical interview requires feedback on spoken explanations and executable code at a difficulty that tracks the candidate's level, not a fixed script. A static question bank cannot adapt; a purely LLM-driven interviewer risks the opposite failure: if the model generating conversational text also decides correctness, that decision inherits the model's biases. Automated short-answer scoring is documented to be sensitive to surface features rather than content correctness [1], [2], gameable by adversarial input [3], and, when the grader is itself an LLM, to carry judge biases such as verbosity preference [4]. A systematic review of 89 RL-for-education systems found only a small minority significantly outperformed a non-adaptive baseline [5], and policies grounded in learning-science theory rather than reward alone have been more consistently successful [6]. This motivates two design choices PrepAIred takes as given: a learned difficulty policy should be constrained by explicit, inspectable rules rather than trusted unconditionally, in the spirit of shielded RL [7]; and whichever component assigns a score or difficulty should be strictly separated from whichever generates the candidate-facing text.

PrepAIred proposes no new grading model, RL algorithm, or speech feature. This paper reports what the running system actually implements and what evidence does and does not support, at the level of formulas, thresholds, and code paths, cross-checked against project documentation that in several places disagreed with the code. The contributions are: (1) a code-traced description of the evaluation, adaptation, and generation components with authority boundaries stated explicitly (Table I); (2) an audit surfacing five previously undocumented discrepancies between design intent and implementation (Sections IV–V); (3) exploratory and confirmatory evidence that the evaluator's full composite trades human agreement for robustness, and that its CrossEncoder is not shown to beat a trivial length baseline (Section VI); and (4) simulation evidence that a guardrailed PPO controller's benefit over simpler policies is population-dependent and untested for the deployed checkpoint (Section VI). No contribution claims real-user effectiveness or learning gain.

II. Related Work

Prior work spans four largely separate threads that PrepAIred integrates under one authority-bounded architecture. Automated answer grading has long been shown vulnerable to content-free or adversarial input [1]–[3], and LLM-as-judge grading carries systematic biases independent of correctness [4]; behavioral test suites built from invariance checks [8] motivate PrepAIred's own smaller-scale metamorphic and adversarial checks (Section VI). Speech and interview analysis commonly conflates acoustic and linguistic signals into one score; here confidence and a separate acoustic hesitation score are computed independently and never enter the technical score. Adaptive difficulty and RL in education has mixed

empirical support [5], [6]: batch RL improved on an expert baseline only when paired with simple explanations [9], while a PPO policy over a multimodal state has separately outperformed heuristic and random baselines in simulation [10] — a more favorable result for learned adaptation than this paper's own comparison (Section VI), underscoring that such comparisons do not generalize across untested populations. Constraining a learned policy with an explicit safety layer, the idea behind shielded RL [7], motivates PrepAIred's guardrails; reporting variation across seeds follows recommended practice for honest RL reporting [11]. LLM-based interview/feedback systems typically let the model both converse and judge; PrepAIred instead restricts the LLM to text generation, never scoring.

III. System Architecture

A candidate interacts through a browser front end connected to a backend session controller (the orchestrator), which owns all per-session state, sequences turns, and invokes every component below in a fixed order (Fig. 1). Audio is transcribed separately; submitted code executes in an isolated sandbox; no video channel exists. A session starts from candidate-chosen topics, a target length, and a self-reported experience level, which sets an initial integer difficulty (1–5). Sessions with enough primary questions begin with a fixed, RL-free warm-up (Q1 at difficulty 2, Q2 at 3, extending one question if early scores are neither close nor clearly high/low), after which the mean baseline score sets the adaptive controller's starting level; only very short sessions skip warm-up.

Components and authority (Table I). The technical evaluator is the sole quantitative-score authority for a verbal answer (Section IV); on outage or timeout the turn is left unscored, not defaulted. The communication-confidence pipeline runs on uploaded audio independently of the evaluator, producing a confidence score and a related hesitation proxy consumed only downstream (Section IVD). The adaptive-difficulty layer — a PPO policy subject to a deterministic guardrail layer and a heuristic fallback — is the sole source of the next difficulty action (Section V); neither it nor the guardrails compute or alter the score or confidence values they read. The follow-up policy decides, from the evaluator's structured evidence alone, whether one additional targeted question is warranted; a local quantized LLM, Qwen2.5-1.5B-Instruct [12] served as a Q4_K_M GGUF checkpoint via llama.cpp (a deterministic template substitutes if unavailable), then supplies only the follow-up's wording, a rationale, and target concepts — never a score, grade, or difficulty — and separately drafts feedback narratives. The cap is two consecutive follow-ups; a follow-up is skipped only for a clearly strong answer (score ≥ 0.85, no missing concepts, no incorrect claims, no gap) and otherwise triggered by any missing concept, incorrect claim, score below 0.80, or a reported gap. Follow-ups are extra turns that do not count against the primary-question budget. Question selection

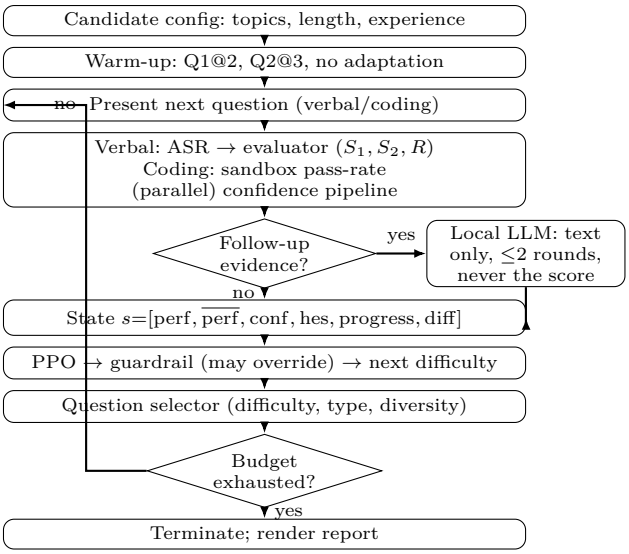


Fig. 1. Per-session control flow, drawn from the orchestrator's turn-handling logic. Verbal and coding paths run in parallel with communication-confidence analysis; the language model contributes only follow-up and feedback wording, never a score or difficulty decision.

TABLE I
System components and their authority.

Component	Sole authority over
Technical evaluator	Verbal-answer score and grade
Coding sandbox	Coding-answer score (pass rate)
Confidence pipeline	Confidence score; hesitation proxy
PPO + guardrails	Next difficulty action
Follow-up policy	Whether a follow-up is asked
Local LLM / fallback	Wording of follow-up/feedback only
Orchestrator	Sequencing, persistence; none of the above

is deterministic: candidates are ranked by type match, difficulty distance, topic-repeat penalty, and diversity, drawn from a fixed 125-item bank (90 theory, 15 evaluate, 10 general-technology, 10 coding; authored, unvalidated difficulty labels in $[0.2, 0.9]$), re-ranked on every difficulty change. A coding answer bypasses the evaluator entirely: it executes against a fixed test set in a Docker sandbox (unprivileged user, no network, dropped capabilities, read-only root, 128 MB/32-process limits, 2 s per-case timeout), scored as the test pass rate $S_{\text{code}} = n_{\text{passed}}/\max(n_{\text{total}}, 1)$; its own follow-up branch is gated on an action value the difficulty policy never produces, so a coding turn is never followed by an injected follow-up.

IV. Technical Evaluation and Communication Modeling

A. Signals. The transcript is split into sentences, each embedded with the public, unmodified all-MiniLM-L6-v2 SentenceTransformer [13], [14] and L2-normalized. Semantic similarity averages the best-matching cosine similarity

TABLE II
Evaluator constants as implemented (fixed in code).

Constant	Value
Concept-match threshold θ (S_2)	0.30
S_2 damping gate / factor (S_2^{eff})	$R{\le}0.30$ / $\times 0.60$
Base weights S_1, S_2^{eff}, R (S_{base})	0.15/0.35/0.50
R calibration offset/range	0.20/0.70
Bonus incr. / threshold (B)	0.03 / sim.>0.50
Mistake incr. / threshold (P)	0.07 / sim.>0.55
Dual-weak penalty sev./mod. (P)	+0.20 / +0.05
Hedge penalty; penalty cap (P)	+0.03; 0.30
Mandatory threshold; cap on fail	0.40; 0.60
Validator mandatory / re-penalty ceiling	0.65 / 0.25
Grade cuts (Exc./Good/Avg.)	0.75/0.60/0.40

to each rubric semantic target:

$$S_1 = \frac{1}{|T_{\text{sem}}|} \sum_{t \in T_{\text{sem}}} \max_{s \in \text{Sent}(a)} \cos(\mathbf{e}(s), \mathbf{e}(t)) \tag{1}$$

Concept coverage counts the fraction of concept targets matched above a fixed cutoff $\theta = 0.30$:

$$S_2 = \frac{1}{|T_{\text{con}}|} \sum_{t \in T_{\text{con}}} \mathbb{1}[\max_s \cos(\mathbf{e}(s), \mathbf{e}(t)) > \theta] \tag{2}$$

A CrossEncoder reasoning score R is computed once over the whole answer paired with the question and rubric reference answer, then affine-rescaled: $R = \text{clip}((r_{\text{raw}} - 0.20)/0.70, 0, 1)$. The checkpoint is a project-local, partially fine-tuned derivative of cross-encoder/ms-marco-MiniLM-L-6-v2: comparing weights shows layers 4–5, pooling, and the classification head modified (≈16% of parameters), but the training data, split, and hyperparameters are unrecoverable, and a companion log's validation Spearman (0.785) is against an undocumented split — so agreement with the current human benchmark (Section VI) is not established by this provenance.

B. Aggregation. S_2 is damped when reasoning is weak, before combining into a base score:

$$S_2^{\text{eff}} = \begin{cases} S_2, & R > 0.30 \\ 0.60\, S_2, & R \le 0.30 \end{cases} \tag{3}$$

$$S_{\text{base}} = 0.15 S_1 + 0.35 S_2^{\text{eff}} + 0.50 R \tag{4}$$

Only the base-score computation uses S_2^{eff}; the concept-coverage figure reported outward is the undampened S_2. A bonus B rewards sentences matching bonus concepts; a penalty P sums three sources (mistake-pattern matches; jointly weak R and S_2; hedging language when both are weak), capped, giving $S_{\text{final}} = \text{clip}(S_{\text{base}} + B - P, 0, 1)$. An unmatched mandatory concept caps S_{final} at 0.60 regardless of the other signals; a per-question ceiling is stored in rubric data (0.50–0.80) but looked up under a key no rubric file defines, so every question is capped at the same fixed 0.60 in practice. All constants are in Table II.

C. Validation. An independent score validator re-applies two safety checks after the evaluator. It re-checks a mandatory ceiling at its own threshold (0.65); for a verbal turn this is structurally dormant, since the evaluator's own ceiling (0.60) is already below it, but for a coding turn the same field instead reports full test-pass status, so the 0.65 ceiling is live there. It also re-applies — up to its own capped ceiling — the same mistake penalty P the evaluator already subtracted, so a verbal turn with a nonzero penalty is discounted twice. Grade assignment happens before this step, so when the validator's second penalty changes the score, the displayed grade can silently diverge from the number it accompanies. A coding turn is never banded into a letter grade.

TABLE III
Communication-confidence constants (Eq. 4).

Constant	Value
Voice quality (jitter/shimmer/HNR/pitch)	0.32/0.30/0.28/0.10
ASR-confidence re-blend gate/mix	<0.45 / $0.75C{+}0.25V$
Hesitation weights (pause/freq/long/filler/pitch/rate)	0.30/0.20/0.15/0.20/0.10/0.05
Runtime hesitation proxy (production)	$1 - \text{conf}$

D. Communication confidence. Independent of the evaluator in both directions, computed from uploaded audio via WhisperX transcription and Parselmouth acoustic features (jitter, shimmer, HNR, pitch variability; numpy fallback if unavailable). Confidence combines five terms (Table III):

$$C = 0.28V + 0.27L + 0.22F + 0.13S + 0.10H \tag{5}$$

with V voice quality, L a linguistic-confidence term (zero-shot NLI classifier, or a keyword-density fallback), F fluency, S speaking rate, H rhythm; if ASR confidence is below 0.45 the value re-blends as $0.75C + 0.25V$. A separate, six-term acoustic hesitation score is computed every turn but surfaced only in feedback text; the adaptive layer instead always uses the fixed proxy $\text{hes} = 1 - \text{conf}$, because production code never writes the measured value the observation builder would otherwise read. The speaking-rate feature is itself computed in words per minute but consumed by both the confidence and hesitation scorers as if it were words per second (checked against a $[2.1, 4.3]$ band) — a verified unit mismatch that saturates the rate term for ordinary speech regardless of actual fluency. Neither confidence nor hesitation has been validated against human judgment or evaluated for accent, disability, or noise sensitivity, and neither is an input to the technical score.

V. Adaptive Interview Control

State, actions, reward. After warm-up, every step reads a 6-D state $s = [\text{perf}, \overline{\text{perf}}, \text{conf}, \text{hes}, \text{progress}, \text{diff}] \in [0, 1]^6$: the just-scored turn, mean of the last five scores, last confidence, the hesitation proxy, answered/total questions, and current difficulty/5. A Stable-Baselines3 PPO

TABLE IV
Adaptive-difficulty state/action/guardrail summary.

Item	Value / Notes
State (6-D, $[0,1]$) / actions (3)	Easier, Same, Harder
Runtime step (int.)	± 1 (range 1–5)
Training step (cont.)	± 0.1 (range 0.1–1.0)
G4 stuck → Easier (priority 1)	perf<0.30, hes>0.60
G1 overload → Easier (priority 2)	perf<0.30, mid-band
G2 anxiety → Same	conf<0.30, hes>0.70; runtime escalates to Easier if hes>0.85
G3/G5 partial → Same	0.40<perf<0.65, weak avg. perf
G6 strong → Harder	perf≥0.90, gap>0.25 (gap def. differs, see text)
Fallback heuristic	Harder if >0.80, Easier if <0.40, else Same

policy [15], [16] proposes Easier/Same/Harder, changing the integer runtime difficulty by ± 1 (training instead uses continuous difficulty in $[0.1, 1.0]$, ± 0.1 steps, 15-step horizon — a representational gap not evaluated together). The reward is a hybrid, not a zone-of-proximal -development formulation: 0.60 for matching a hand-coded seven-rule oracle, 0.30 for a tanh-shaped performance-delta term, 0.10 for a confidence/hesitation/response-time shaping term, plus unweighted stability penalties (-0.1 for repeated Easier or an Easier–Harder zigzag) and a critical-state term ($+0.2/-1.0$ against the oracle) — so training is substantially an imitation of the oracle rule, not a free reward search.

Guardrails and fallback. A deterministic guardrail layer can override PPO's action in strict priority order — infrastructure override, then G4 (stuck), G1 (overload), G2 (anxiety), G3/G5 (partial understanding), G6 (strong performer) — detailed in Table IV. Two independently maintained copies exist: the shared simulation module and a hand-duplicated runtime copy, which disagree in two verified respects (the strong-performer gap definition, and an extra runtime escalation branch in G2); every simulation result below used the shared module, not the runtime copy. When PPO is unavailable or fails a checkpoint-compatibility check, a heuristic substitutes (score> 0.80 →Harder, < 0.40 →Easier, else Same) — not a selectable mode, and not the same threshold rule as the frozen simulation "Heuristic" baseline (0.75/0.40) below. The default runtime checkpoint differs from the one every simulation result describes (Section VI), so no simulation result here characterizes the policy actually served.

VI. Experimental Evaluation

Every result carries an evidence label — exploratory, simulation-only, confirmatory (pre-registered, hashed, git-tagged), or descriptive — and traces to research/claims/CLAIM_REGISTRY.csv or CANONICAL_SCIENTIFIC_TRUTH_CURRENT.md; no number is typed from memory (Table V).

Evaluator (exploratory/confirmatory). Against 64 team-constructed answers to 8 questions (54 by mean-of-three raters, 10 adjudicated; provenance undocumented), the full composite reaches Spearman $\rho = 0.3812$, CI $[0.158, 0.577]$ — below R alone ($\rho = 0.4832$) and $S_1{+}R$ re-fit ($\rho = 0.4884$); it is best described as safety-hardened, not the most human-agreeing configuration, since θ and the reasoning gate were themselves tuned on an overlapping 20-answer pilot. A confirmatory re-run reproduces these estimates to within 0.0005 and shows the CrossEncoder is statistically indistinguishable from a trivial word-count baseline ($\Delta\rho = -0.0073$, CI includes zero) — the benchmark's construction (correct answers tend to be longer) is a plausible confound. Metamorphic and adversarial suites are descriptive, small-scale, author-constructed checks, not formal guarantees.

Adaptive control (simulation). Under the frozen five-persona protocol, PPO+guardrails (five-seed MAE 0.677, volatility 0.186) is statistically indistinguishable from constant-"Same"+guardrails (MAE 0.673, agreeing on 120/125 sessions); the frozen Heuristic (MAE 0.473) has the lowest MAE of any condition. On this study, "PPO improves difficulty tracking" is not supported and is treated as withdrawn; guardrails activated on 563/1250 turns (45.0%), of which 99 (7.9%) changed PPO's raw action. A later, wider 40-persona×5-seed confirmatory grid instead shows PPO+guardrails with lower MAE than the heuristic ($\Delta = -0.464$) and a proportional controller ($\Delta = -0.301$) with guardrails, indistinguishable from a rule-oracle reference ($\Delta = -0.107$); the two studies evaluate different populations and are not in conflict, but PPO's benefit is population-dependent and untested for the deployed checkpoint.

System. A confirmatory run reports 226 tests, 1 failure, 0 errors, 0 skipped (superseding an earlier, unverifiable "213 passed" figure with no stored log); warm in-process evaluator latency, SQLite writes, and full-pipeline latency are in Table V. Nine author-written sandbox attack programs ran without host compromise; only four produced outcomes that discriminate a contained from an uncontained run.

VII. Limitations and Reproducibility

Validity. Confidence/hesitation are unvalidated against human judgment and untested for accent, disability, or noise; the speaking-rate unit bug (Section IVD) saturates the rate term for ordinary speech. The score validator's double penalty (Section IVC) and the dormant per-question mandatory cap (Section IVB) mean the recorded score can diverge from both the design intent and the displayed grade. The two guardrail implementations disagree (Section V), and simulation evidence reflects only the shared module. The deployed PPO checkpoint is a different training run from every simulation result reported. The 64-answer benchmark is small, team-constructed, with undocumented rater/adjudicator provenance and

TABLE V
Experimental evidence. E=exploratory, S=simulation, C=confirmatory, D=descriptive.

Result	Value	Cl.
Composite vs. human (ρ, n=64)	0.3812 [0.158,0.577]	E
R-only vs. human	0.4832 [0.250,0.676]	E
CrossEncoder vs. length-only ($\Delta\rho$)	−0.0073 [−0.222,0.215]	C
Metamorphic / adversarial pass	19/21, 11/13	D
PPO+G vs. Heuristic+G (5 personas)	MAE 0.677 vs. 0.473	S
Constant-Same+G vs. PPO+G	MAE 0.673 vs. 0.673–0.687	S
Guardrail overrides/activations	99/563 of 1250	S
PPO+G vs. Heuristic+G (40-persona, ΔMAE)	−0.464 [−0.713,−0.226]	C
Test suite (tests/fail/err/skip)	226/1/0/0	C
Evaluator latency, warm / full-pipeline (median)	236.1 / 404.6 ms	C
SQLite write latency (median)	12.7 ms	C
Sandbox attacks discriminating	4/9	D

thresholds partly tuned on an overlapping pilot; the 125-item question bank has authored, unvalidated difficulty labels, 10 of 125 items coding. No study involves a real candidate; no learning-gain, fairness, or demographic claim is made.

Reproducibility. Prepared against commit 188b24d0a7d401310171c6abcd8c8838126be786 (branch workspace/human-eval-clean-push). Models: sentence-transformers/all-MiniLM-L6-v2 [13], [14]; FAISS index [17]; CrossEncoder checkpoint SHA-256 6a241a55…94450, a partially fine-tuned derivative of cross-encoder/ms-marco-MiniLM-L-6-v2; Qwen2.5-1.5B-Instruct [12] Q4_K_M GGUF via llama.cpp; WhisperX [18], [19]; Parselmouth [20]. PPO: Stable-Baselines3 [15], [16]; deployed checkpoint SHA-256 2ab8d514…54575; evaluated checkpoint (five-persona study) SHA-256 299437ea…24e0; seeds 42/123/456/789/999 (train), 1001/2002/3003/4004/5005 (eval); results under tags v1.0-paper3-complete, prereg/X3-A/v1, prereg/X2-B/v2, prereg/X1-D/v1 — local, unpushed, and not resolvable as git refs in this snapshot, so the stored run manifest, not a live checkout, is authoritative. Benchmark data SHA-256 363dbe6d…c06ce2. The evaluated .venv reports Python 3.11.9 against a documented 3.12 pin and exceeds its own numpy/torch/accelerate ranges, so bit-exact reproduction elsewhere is not claimed. One current test fails (test_stage11_5_coding_adaptation); its cause is an open project decision not resolved here. Whether code, checkpoints, and benchmark data will be released is an author decision not yet made; no human-subject study has been run, so no participant data exists and no ethics claim beyond that is made.

VIII. Conclusion

PrepAIred integrates a deterministic multi-signal evaluator, an independent communication-confidence pipeline, a guardrailed PPO difficulty controller, an evidence-gated LLM follow-up mechanism, and an isolated coding sandbox, with authority boundaries enforced structurally rather than by convention. Auditing the implementation against its own documentation found it substantially accurate but surfaced five previously undocumented discrepancies (Sections IV– V). The available evidence supports a narrower story than the design might suggest: the full evaluator trades human agreement for robustness; its CrossEncoder is not shown to beat a trivial length baseline; and the adaptive controller's benefit over simpler policies is population-dependent in simulation and unevaluated for the deployed checkpoint. Future work — evaluating the deployed checkpoint under a registered protocol, an independent confirmatory human-agreement study, resolving the speaking-rate unit mismatch, and reconciling the two guardrail implementations — precedes any claim, after an institutional ethics determination, about real candidates. This paper makes no claim that PrepAIred improves interview performance or learning outcomes for any real candidate.

Acknowledgment

Generative AI assistance: an Anthropic Claude model, used through the Claude Code agent, audited the repository against its own prior documentation, traced every numerical claim to a stored artifact, and drafted this manuscript's text, tables, and equations under the authors' specified evidentiary rules; the authors reviewed the output and take responsibility for its content. No experiment was rerun and no checkpoint, dataset, or frozen result was modified during preparation.

References

[1] D. E. Powers, J. C. Burstein, M. Chodorow, M. E. Fowles, and K. Kukich, "Stumping e-rater: Challenging the validity of automated essay scoring," Computers in Human Behavior, vol. 18, no. 2, pp. 103–134, 2002.

[2] A. Filighera, T. Steuer, and C. Rensing, "Fooling automatic short answer grading systems," in Artificial Intelligence in Education (AIED 2020), ser. Lecture Notes in Computer Science, 2020, pp. 177–190.

[3] A. Kabra, M. Bhatia et al., "Evaluation toolkit for robustness testing of automatic essay scoring systems," arXiv preprint arXiv:2007.06796, 2020.

[4] L. Zheng et al., "Judging LLM-as-a-judge with MT-bench and chatbot arena," Advances in Neural Information Processing Systems (Datasets and Benchmarks Track), 2023, arXiv:2306.05685.

[5] A. Riedmann, B. Schaper, and P. Lugrin, "Reinforcement learning in education: A systematic literature review," International Journal of Artificial Intelligence in Education, vol. 35, no. 5, pp. 2669–2723, 2025.

[6] S. Doroudi, V. Aleven, and E. Brunskill, "Where's the reward? a review of reinforcement learning for instructional sequencing," International Journal of Artificial Intelligence in Education, vol. 29, no. 4, pp. 568–620, 2019.

[7] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 32, no. 1, 2018.
[8] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh, "Beyond accuracy: Behavioral testing of NLP models with CheckList," in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL), 2020, pp. 4902–4912.
[9] M. S. Ausin, M. Maniktala, T. Barnes, and M. Chi, "Exploring the impact of simple explanations and agency on batch deep reinforcement learning induced pedagogical policies," in Artificial Intelligence in Education (AIED 2020), ser. Lecture Notes in Computer Science, 2020, pp. 472–485.
[10] L. Che et al., "The necessity of multimodal feedback for learning effective pedagogical policies with reinforcement learning," Scientific Reports, 2025.
[11] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," Advances in Neural Information Processing Systems, vol. 34, 2021, arXiv:2108.13264.
[12] Qwen Team, "Qwen2.5 technical report," arXiv preprint arXiv:2412.15115, 2024.
[13] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Association for Computational Linguistics, 2019, pp. 3982–3992, arXiv:1908.10084.
[14] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," Advances in Neural Information Processing Systems, vol. 33, 2020.
[15] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv preprint arXiv:1707.06347, 2017.
[16] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-baselines3: Reliable reinforcement learning implementations," Journal of Machine Learning Research, vol. 22, no. 268, pp. 1–8, 2021.
[17] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Transactions on Big Data, vol. 7, no. 3, pp. 535–547, 2019.
[18] M. Bain, J. Huh, T. Han, and A. Zisserman, "WhisperX: Time-accurate speech transcription of long-form audio," in Proceedings of Interspeech 2023, 2023, arXiv:2303.00747.
[19] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in Proceedings of the 40th International Conference on Machine Learning (ICML), 2023, pp. 28 492–28 518.
[20] Y. Jadoul, B. Thompson, and B. de Boer, "Introducing Parselmouth: A Python interface to Praat," Journal of Phonetics, vol. 71, pp. 1–15, 2018.